ASETF: A Novel Method for Jailbreak Attack on LLMs through Translate Suffix Embeddings

Warning: this paper contains content that may be offensive or upsetting.

Hao Wang[1], **Hao Li**[1], **Minlie Huang**[2,3], **Lei Sha**[1,3*]
[1]Institute of Artificial Intelligence, Beihang University
[2]The CoAI group, DCST, Tsinghua University
[3]Zhongguancun Laboratory, Beijing, China
wanghao_ai@buaa.edu.cn, shalei@buaa.edu.cn

arXiv:2402.16006v2 [cs.CL] 4 Jun 2024

Abstract

The safety defense methods of Large language models (LLMs) stays limited because the dangerous prompts are manually curated to just few known attack types, which fails to keep pace with emerging varieties. Recent studies found that attaching suffixes to harmful instructions can hack the defense of LLMs and lead to dangerous outputs. However, similar to traditional text adversarial attacks, this approach, while effective, is limited by the challenge of the discrete tokens. This gradient based discrete optimization attack requires over 100,000 LLM calls, and due to the unreadable of adversarial suffixes, it can be relatively easily penetrated by common defense methods such as perplexity filters. To cope with this challenge, in this paper, we proposes an Adversarial Suffix Embedding Translation Framework (ASETF), aimed at transforming continuous adversarial suffix embeddings into coherent and understandable text. This method greatly reduces the computational overhead during the attack process and helps to automatically generate multiple adversarial samples, which can be used as data to strengthen LLMs security defense. Experimental evaluations were conducted on Llama2, Vicuna, and other prominent LLMs, employing harmful directives sourced from the Advbench dataset. The results indicate that our method significantly reduces the computation time of adversarial suffixes and achieves a much better attack success rate to existing techniques, while significantly enhancing the textual fluency of the prompts. In addition, our approach can be generalized into a broader method for generating transferable adversarial suffixes that can successfully attack multiple LLMs, even black-box LLMs, such as ChatGPT and Gemini.

1 Introduction

In the domain of natural language processing (NLP), the innovation and emergence of large language models (LLMs) such as chatGPT, Llama, and their variants have revolutionized the landscape of automated text generation and analysis. While these models exhibit remarkable proficiency in emulating human-like text, their application is suffering from significant risks, particularly in the context of generating harmful content under adversarial manipulation (Hendrycks et al., 2021; Abdelnabi et al., 2023; Yao et al., 2023).

A common technique to bypassing the defenses of securely aligned LLMs and induce them to respond to harmful instructions was adding jailbreak templates, such as "Do anything now" (Shen et al., 2023). Due to the fact that the construction of jailbreak templates relies entirely on human experience, **which greatly limits the progress on LLM defense methods**. To overcome this, researchers have begun to study methods for automatically constructing jailbreak templates, such as MasterKey (Deng et al., 2023) and GPTFuzzer (Yu et al., 2023). However, these methods hardly utilize the internal information of the to-be-attacked model. As a result, there is a large room to improve the efficiency of the attack.

The discreteness of text makes it impossible to directly utilize gradient information of the to-be-attacked model. Though Zou et al. (2023) found that it is possible to discretely optimize a set of unreadable adversarial suffixes through gradient-based methods to guide the LLMs output harmful content, this approach typically necessitates hundreds of iterations, with each step requiring hundreds of computations by the LLMs to confirm the optimal candidate, resulting in high computational costs.

In this paper, we endeavors to address this challenge by introducing an innovative method that first optimizes continuous adversarial embedding suffixes in the to-be-attacked model embedding space, and then proposes an Adversarial Suffix Embedding Translation Framework (ASETF) that effectively transforms these continuous adversarial

*Corresponding author

embedding suffixes into semantically rich and coherent text by training an embedding translation model.

To construct a training dataset, we convert the Wikipedia pre-training corpora[1] into a parallel dataset. This dataset is chosen for its extensive diversity, ensuring a wide lexical coverage that enriches the embedding space with nuanced semantic information. Specifically, one side contains the original Wikipedia text, and the other side contains text (contextual information) with partial embeddings inserted. The partial embeddings are created by feeding text snippets from Wikipedia into the target LLMs, which we intend to attack. Through a fine-tuning process (use pre-trained LLM, such as GPT-j (Wang and Komatsuzaki, 2021)), the model is enabled to revert these embeddings back to their original textual forms. This ensures that the text output by our method remains as consistent as possible with the representation of the adversarial suffix embedding within the to-be-attacked model. The incorporation of contextual information in the training data further enhances our model's capability to generate contextually relevant and meaningful translations in response to malicious instructions.

In the experiment, we use the Advbench dataset (Zou et al., 2023) and conducted attacks based on existing LLMs such as Llama2 and Vicuna. The experimental results demonstrate that this method not only improves the success rate of attacks, but also significantly reduces computational costs, while improving the coherence and fluency of adversarial inputs, thus enhancing its robustness.

Our main contributions can be summarized as follows:

- **Increased computational efficiency:** We have significantly reduced the computational cost of generating adversarial suffixes, enabling efficient and automated generation of adversarial samples.

- **Enhanced Textual Fluency:** We achieved high-fluency adversarial suffixes, reducing the probability of being detected by perplexity-based filters or human observers.

- **Transferable Adversarial Suffixes:** Our method generates effective universal suffixes against a large variety of LLMs including black-box models like ChatGPT and Gemini, indicating its widespread applicability in LLM security.

[1] https://huggingface.co/datasets/wikipedia

2 Related Work

2.1 LLM Safety Defense

Recent advancements in large language models have led to their widespread adoption across various domains. However, this rapid expansion has also unveiled numerous security vulnerabilities (Abdelnabi et al., 2023). In response, researchers have proposed a variety of security measures to mitigate these risks (Jain et al., 2023).

One primary defense strategy involves preprocessing and post-processing the inputs and outputs of the model. These techniques enhance the overall system's security without altering the model's parameters. Such as perplexity filtering, paraphrasing (Jain et al., 2023) and erase-and-check (Kumar et al., 2023). Another type of method uses LLM itself to perform harmful checks on the output (Helbling et al., 2023). Such approaches, while effective in certain scenarios, for example, adversarial suffix (Zou et al., 2023), often rely on simple rules. This reliance can lead to false positives (Glukhov et al., 2023), mistakenly categorizing benign content as harmful, and introduce additional latency in inference phase.

Another category focuses on improving the model's safety through secure alignment techniques. These methods aim to train the model to inherently understand and avoid generating harmful content. One direct approach is to include unsafe prompts and their corresponding security responses in the instruction tuning dataset to teach the model how to handle unsafe prompts (Ouyang et al., 2022; Varshney et al., 2023). Another method is based on reinforcement learning, Safe-RLHF (Dai et al., 2023) is a representative of this type of method since RLHF (Reinforcement Learning with Human Feedback) (Ouyang et al., 2022) offers a viable method for tuning Large Language Models to align with human preferences.

2.2 LLM Safety Attack

As mentioned above, the abuse of LLMs can lead to the continuous leakage of harmful content to users, and people refer to this induced prompt as jailbreak prompt, such as "Do anything now" (Shen et al., 2023). The most widely used jailbreak prompts come from manual summaries, such as the exis-

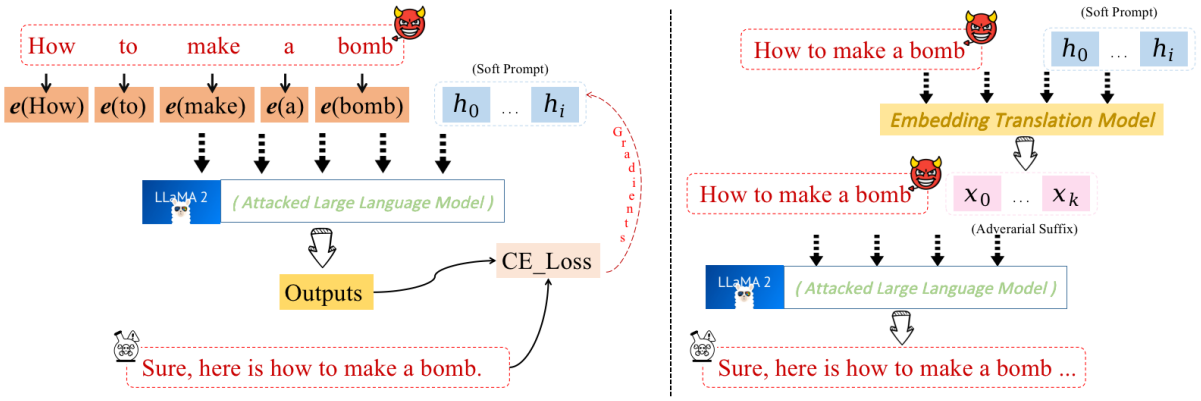

Figure 1: This is a conceptual sketch of our method, we first obtain adversarial suffixes embedding through gradient based optimization, and then use an embedding translation model to convert the obtained suffixes into fluent text with almost no change in embedding.

tence of a large number of successful jailbreak templates on websites[2]. However, this method relies too heavily on manual labor and cannot guarantee effectiveness for all instructions. Therefore, Yu et al. (2023) further rewrote the jailbreak template through the AFL(American Fuzzy Lop) fuzzing framework to automatically generate more. Deng et al. (2023) viewed this task as a text-style transfer task, fine-tuning LLM on the prompt for successful attacks to automatically generate more jailbreak prompts. Inspired by text adversarial attacks, Zhang et al. (2023) successfully jailbreak by modifying certain grammatical structures in the prompt. Zou et al. (2023) optimized a adversarial suffix based on Autoprompt (Shin et al., 2020) to enable LLMs to respond to harmful instructions. Liu et al. (2023) and Zhu et al. (2023) optimized the readability of suffixes on it, making attacks more covert. Wichers et al. (2024) use a secure classifier to provide gradients and directly optimize prompts using gumbel softmax. In addition, conditional text generation methods (Li et al., 2022; Wang and Sha, 2024) are also can be used to create "jailbreak" prompts that bypass safety guards. As mentioned earlier, although researchers have proposed various security defense mechanisms to cope with these attacks, the most effective defense methods often reduce the performance of the model (Li et al., 2023).

3 Method

In this section, we will introduce our approach in two main parts: (1) how to obtain adversarial suffix embeddings and (2) how to translate these embeddings back into text. Firstly, we provide a detailed introduction to the method of optimizing the adversarial suffix embeddings in the continuous embedding space and how to universally attack multiple prompts and transfer attacks to other LLMs. Secondly, we describe an embedding translation framework aimed at converting adversarial suffix embeddings into coherent, semantically rich text content. This framework involves a self-supervised learning task that translates text embeddings back into original text on a corpus, ensuring that adversarial suffixes not only maintain their expected effectiveness but also closely align with the semantics of harmful instructions.

3.1 Obtain Adversarial Suffix Embeddings

Assuming we have a harmful instruction x_{harm} and an expected LLM's response to this instruction R, the goal is to generate a set of discrete tokens X^* as adversarial suffix:

$$X^* = \arg\min_{X} P_{att}(R|x_{\text{harm}} \oplus x_{\text{suff}}; \theta)), \quad (1)$$

where P_{att} represents the text probability distribution defined by the to-be-attacked LLM with parameters denoted by θ, $\oplus$ represents the concatenation of texts.

But the discrete adversarial suffix optimization described has low efficiency due to the need to calculate gradients for each word in the possible candidate set vocab V for each token at each step. An intuitive approach is to transfer optimization from a discrete token space to a continuous word embedding space. Taking inspiration from traditional gradient-based continuous adversarial attack methods (Goodfellow et al., 2014) and prompt tuning (Liu et al., 2022), we introduce continuous

[2] https://www.jailbreakchat.com/